Automating Parcel-Based Co-occurrence Models for Use in Natural Resource Inventories

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Practicum Overview

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Abstract

This study served to automate Natural Resource Inventory parcel-based co-occurrence analyses using Python for integration into ArcGIS. The overall goal was to provide Moosewood Ecological, LLC with a Graphical User Interface (GUI)-based toolset for conducting these analyses for towns in New Hampshire. Four script tools were created: Agricultural and Forest Resources Co-occurrence, Water Resources Co-occurrence, Ecological Resources Co-occurrence, and a tool that weights and combines the previous three outputs. The three co-occurrence tool outputs were cross-validated with co-occurrence analyses conducted manually for the town of Bow, NH. For the automated Agricultural and Forest Resources Co-occurrence analysis, parcel scores deviated from the manual-calculation method by an average of .025 points (.186% difference). The Water Resources Co-occurrence and Ecological Resources Co-occurrence varied from the manual method by an average of 3.67 (49.4% difference) and 1.73 points (31.84% difference), respectively. By automating co-occurrence analyses, Antioch Spatial Analysis Lab (ASAL) can aid Moosewood Ecological, conservation commissions, municipalities, state departments, and other stakeholders in identifying Conservation Focal Areas (CFA) within municipalities and regions of New Hampshire, efficiently. The automated methods may reduce the time to complete the analysis by up to 98.33%. Automation serves to reduce compounding user errors associated with manual-calculation method; the main factor contributing to point-divergence between the two methods.

1. Introduction

1.1 Project and Client Overview

Natural Resource Inventories (NRI) are comprehensive assessments that compile and describe naturally occurring resources within a given area. These areas often follow political jurisdictions such as a municipality, state, or region. NRI can include information regarding cultural resources, such as historic and recreational elements within the area as well [1]. NRI are mandated by New Hampshire statutes for all communities within the state, and this is the responsibility of the town conservation commission [2]. NRI can aid in updating a town's master plan, creating or updating a conservation plan, or helping with the review of new development projects. Through each of these actions, NRI inform land management decisions that may have impacts to citizens (zoning, recreation) and the environment (biodiversity) [1].

NRI tend to include meaningful displays of the data compiled within a report document. These displays can include: maps, charts, tables, and other figures used to describe the resources available in the area. Maps produced in Geographic Information Systems (GIS) are used frequently in these reports to display multiple datasets occurring within the study area. Along with these simple maps, analyses are conducted in GIS and other spatial software packages to provide additional information to a municipality or group.

Moosewood Ecological, LLC is an ecological consulting company based out of Chesterfield, New Hampshire with over 12 years of experience in ecological research and environmental education. NRI and GIS mapping services are two of the many services that Moosewood Ecological offers to its clients within New Hampshire. Moosewood Ecological's owner, Jeffry Littleton, often employs students at Antioch University New England to partake in the mapping and analyses associated with these NRI projects. Specifically, Jeffry works intimately with ASAL to conduct NRI mapping using existing datasets on the NH GRANIT data repository as well as manually-created, ground-truthed datasets generated through field studies.

Often, following the completion of the NRI maps, ASAL conducts a co-occurrence analysis by parcel for the given client or municipality. Co-occurrence is an overlay analysis that weights the presence or percentage of a given feature within the parcel. By scoring the various attributes intersecting a parcel, Moosewood Ecological can assign parcels a relative score in terms of conservation value to the client. The final co-occurrence model can be used to identify CFA so that the client may allocate resources to protecting these ecologically valuable lands [2].

The co-occurrence analysis is a robust, time-consuming analysis that can take an experienced GIS technician roughly 20 to 40 hours to complete for a given area depending on the number, size, and type of datasets used. ASAL, being a close partner of Moosewood Ecological, seeks to reduce the time required to complete the analysis to improve project completion efficiency and increase the capacity of work that can be accomplished by its members. Therefore, the task at hand was to automate the co-occurrence analysis workflow using ArcGIS and Python [3].

Automating co-occurrence mapping for NRI allows ASAL to provide Moosewood Ecological with analysis results quickly and more efficiently than performing the geoprocessing tasks manually. Ultimately, creating a Python script and ArcMap toolset for this analysis, Moosewood Ecological is able to identify areas where natural, cultural, and agricultural resources overlap. This aids Moosewood Ecological's clients in making management and conservation decisions regarding the aforementioned resources. By using Python, this analysis is compatible on any computer device and the results can be shared more readily between partners, provided they also have ESRI ArcGIS software installed. By using GUI-based toolsets, even unsavvy GIS users can perform this analysis. GIS users can perform the analysis for many towns or large swaths of area, providing results for several projects. Alternatively, the tool can be run in multiple instances, producing results for several projects simultaneously. The effectiveness of the tools and their usage is largely dependent on data availability for the given study area.

Using Python and Geographic Information Systems (GIS) software, scripts were generated that calculate the co-occurrence of natural resources, by parcel, on a given landscape. Open source test data was downloaded from NH GRANIT and other sources, transformed, and analyzed using ArcMap 10.7 [4]. PythonWin (a Python IDE), developed by Mark Hammond under the Python for Windows (Win32) extensions was used to script and de-bug the parcel-based co-occurrence models. The scripts were used to transform data and automate geoprocessing tasks. Within ArcMap, a standalone toolset can be added, allowing users of the software to perform the analyses with a dialogue box or graphical user interface (GUI). The dialogue box provides prompts for user input datasets and offers help documentation.

1.2 Co-occurrence Analysis for NRI

Co-occurrence analysis refers to a simple overlay of attributes within a given area of land (e.g., tax parcel). Attributes are aggregated based on presence or absence of feature(s) within zones. However, there are various methods for overlaying and assessing these attributes in terms of conservation value. Since the 1970s and 1980s there has been an expansion of methods and models pertaining to conservation planning. More explicitly, there has been a proliferation of spatial assessment tools and software (e.g., C-Plan, MARXAN and Zonation). As with all ecological assessments and models, each methodology has its own strengths and limitations [5].

Within the conservation planning framework, these tools and models fall into two categories: complementarity-based approaches and scoring-based approaches. Complementarity-based approaches utilize mathematical optimization problems to assess conservation priority [6]. Complementarity approaches often require extensive data input and traditionally have a steep learning curve to use. These methods and tools are often esoteric to local planners and policymakers, making them a poor match for natural resource planning initiatives [5]. In contrast, scoring-based approaches evaluate CFA by using individual scores for multiple attributes and select the top-ranked or top-scored sites from the list of zones (tax parcels). These approaches are often additive and weight individual factors based on expert opinion to create a composite score index [7-9]. Scoring approaches tend to be easier to implement and can be created and understood by a user with moderate GIS skills. Scoring methods are advantageous in that factors can be added or removed easily, making the model flexible [5]. As such, a scoring-based approach was used for this co-occurrence analysis framework.

2. Methods

2.1. Study Area

GIS analysis was performed for the town of Bow, NH, located in Merrimack County. The Town of Bow covers 28.42 square miles, of which 3,000 acres is Town-owned lands and more than 2,200 acres is town forest [10]. In Bow, 16 percent (2,936 acres) of the total land area is identified as Highest Ranked Wildlife Habitat in NH. In addition, approximately 17 percent (3,182 acres) of the total land area is identified as Highest Ranked Habitat in the Biological Region [11]. Results of the Town's Community Survey and Visioning Session conducted as part of the 2017 Master Plan update indicate that stewardship of natural resources is seen as an important investment to maintain the Town's rural quality [10,12]. Therefore, the Town of Bow put forward a call for proposals regarding conducting an NRI [10]. The goals of the NRI were as follows:

- Better management of Town-owned land for wildlife and recreation, including land currently protected
- Identification of Town-owned land that may warrant protection by easements or other means
- Identification of additional land that may warrant protection based on significant natural resource value
- Identification of threats to resources to inform parcel-based land use decisions or changes to current land use regulations
- Support outreach to citizens about the importance of the Town's natural resources.

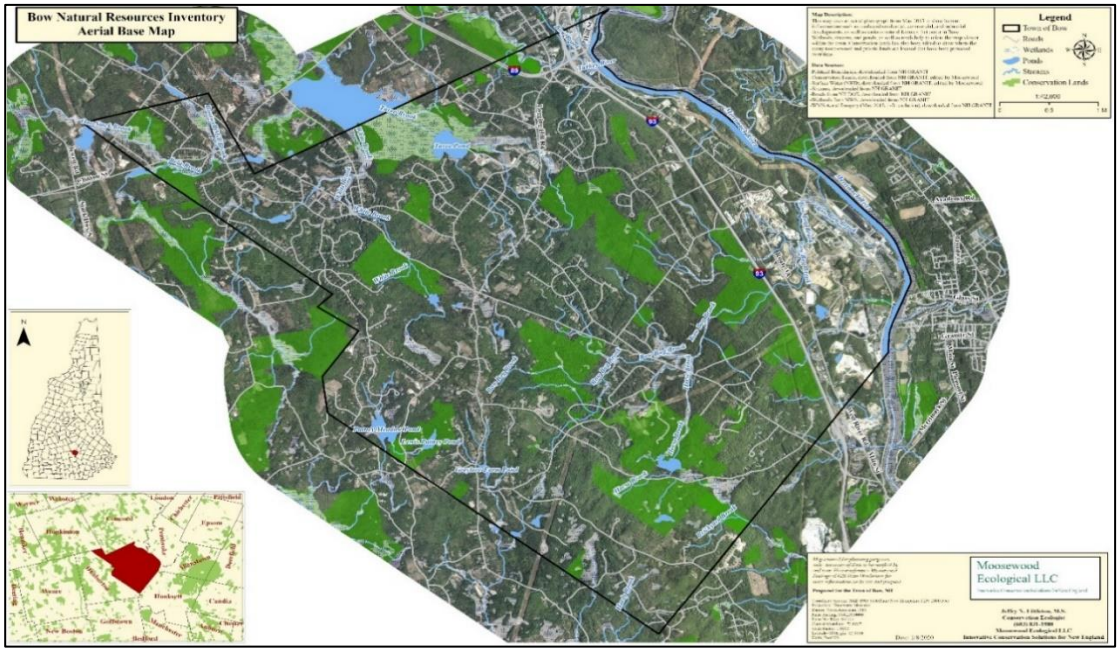

Figure 1: Aerial Base Map for the Town of Bow, NH. Map created by ASAL for Moosewood Ecological, LLC and the Bow Conservation Commission.

2.2 Data

Table 1. Data sources used in the co-occurrence scripts and GIS analyses.

Dataset	Data Source	Year
Active Farmland	Moosewood Ecological	2019
Soils (from SSURGO)	Natural Resource Conservation Service	2009
Rare/Unique Natural Communities	Moosewood Ecological	2019
Forest Cover (NH Land Cover Assessment)	NH GRANIT	2001
Wildlife Action Plan (WAP Habitat and WAP Tiers)	NH GRANIT	2015
Merrimack Conservation Plan	Merrimack Conservation Partnership	2014
Unfragmented Land Blocks	ASAL (generated from NH DOT Roads data)	2013
NH Conservation and Public Lands	NH GRANIT, edited by Moosewood Ecological	2013
NH Parcel Mosaic	NH GRANIT	2019
Rare Species Habitat	Natural Heritage Bureau	2019
National Wetlands Inventory Plus	NH GRANIT	2014
National Hydrography Dataset	NH GRANIT	2005
Stratified Drift Aquifers	NH GRANIT	2000
Favorable Gravel Wells	NH Department of Environmental Services	2010
Wellhead Protection Areas	NH Department of Environmental Services	2013
NH Designated River Corridor	Moosewood Ecological	N/A
Rare/Declining/Vulnerable Habitats	Moosewood Ecological	N/A

2.3 Models

Three co-occurrence model scripts were produced in Python and added to a toolbox named "CoOccurrence.tbx" in ArcMap. The three co-occurrence tools are as follows: Agricultural Forest Resources Co-occurrence, Water Resources Co-occurrence, and Ecological Resources Co-occurrence. A fourth and final Conservation Focal Area script tool was produced which takes the other three co-occurrence outputs as input and performs a weighted overlay (sum) of these datasets to produce a CFA feature class.

These models all utilize three main principles for scoring attributes within a parcel. Percentage overlap of a feature within the parcel (e.g. percent wetland coverage), presence/absence of features (e.g. vernal pools), and number of unique types of an attribute intersecting (e.g. rare species habitats). Based on the scoring criteria provided by Moosewood Ecological (Appendix A), scores were assigned to each parcel based on specified percent thresholds, presence/absence of a feature, or the number of unique features overlapping a parcel. Once all datasets were scored appropriately, they were summed to yield the final conservation value score per parcel.

Figure 2. Toolbox structure for the co-occurrence and conservation focal area tools.

2.3.1 Data Pre-processing and Tool Creation

All datasets except for the town boundary file and parcel layer should be unclipped to the study area. Datasets downloaded from NH GRANIT must contain the original table fields as when they are downloaded from the ftp site. Several datasets for the co-occurrence scripts require special attention when undergoing pre-processing. For instance, hydric soils and forest soils should be extracted (select by attribute) from the soils layer before undergoing further analysis and being entered into the script tools. The tools provide help documentation in the side pane that will guide users through the data requirements. All data inputs must be geodatabase feature classes, not shapefiles.

The procedure for creating all the co-occurrence models was the same. Models were scripted in PythonWin and added to a script tool within the "CoOccurrence" toolbox in ArcCatalog.

Within the GUI, user input parameters were specified that allow a user to input their desired datasets. All of the ecological dataset inputs are optional. The only required parameters (marked by green dots) for the three co-occurrence models are as follows: Current Workspace, Output Directory, Study Area Boundary File, and Parcel Layer Input. The Current Workspace refers to the geodatabase where all of the input datasets are located. It is suggested that a separate input geodatabase is created for each co-occurrence model. Output Directory refers to the location where the user would like the intermediate files and final co-occurrence feature class to reside. Study Area Boundary File refers to the boundary polygon file of your study area that all the subsequent datasets will be clipped to. Parcel Layer Input refers to the New Hampshire Parcel Mosaic from NH GRANIT. All other layers are optional and the optional dataset inputs vary depending on which co-occurrence script is being run. For the Conservation Focal Areas tool, only the Current Workspace and Output Directory are required.

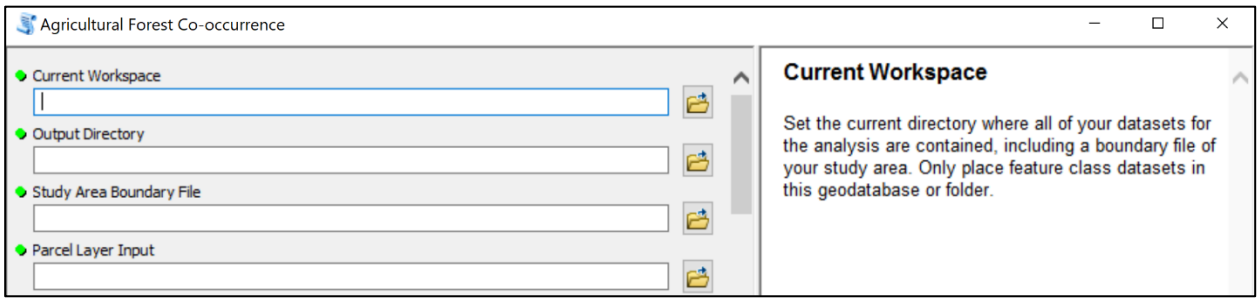

Figure 3. Example of required parameters from the Forest Agricultural Co-Occurrence Analysis GUI.

2.3.2 Agricultural and Forest Resources Co-occurrence Model

The Agricultural and Forest Resources Co-occurrence is a parcel-based GIS assessment model that provides a sense of the relative agricultural and forest resources value for each parcel in the study area. This model serves as a tool that helps conservation commissions and other town boards to identify conservation priorities. The model provides a basis from which town(s) may proceed with conservation projects with willing landowners that are interested in protecting their property. This assessment model ranks all parcels in the same manner and is based on a cumulative point system, providing a final overall score for each parcel. This affords the opportunity to directly compare one parcel with another. The assessment for each parcel includes many attributes: forest and agricultural soil types, active farmlands, percent forest cover, rare/unique forest communities, regional conservation plans, size of the property, fragmentation, proximity to conserved lands, and rare species occurrences.

i. Layers for Percent Intersect

Foregoing the required datasets (Section *2.3.1*), the tool then prompts users to input Layers for Percent Intersect. These are the feature classes that will be scored based on their percentage overlap with a parcel. For this particular model this may include: active farmland, farmland soils, forest soils, and forest cover. More datasets can be specified if needed. Within the GUI, multiple datasets can be input into this parameter and the feature classes will be parsed separately within the script. For each of these feature classes, the analysis is the same. The script tool takes each input dataset, clips it to the study boundary file, and runs the Tabulate Intersect tool. The zones for the tool consist of a unique Parcel FID assigned to each feature in the parcels dataset. This tool computes the intersection between two feature classes and cross-tabulates the area, length, or count of the intersecting features within the parcel. For this analysis, the percent of areal overlap is calculated in the Percentage field of the script tool's output table (Figure 4).

ClippedBow_AgSoilsTabulateIntersect

OBJECTID *	Parcel_FID	AREA	PERCENTAGE
1	1	3038.391433	100
2	2	5124.609168	100.000947
3	3	59.444247	100
4	4	5917.690673	100.00056
5	5	33848.185515	100
6	6	32132.762957	100
7	7	1471.698587	100
8	8	284482.511701	99.999938
9	9	106340.645549	99.999965

Figure 4. Example of the Tabulate Intersect output table for agricultural soils.

Once Tabulate Intersect is run, a score field is added to the output table. This Tabulate Intersect table is then joined back to the parcels layer input using the Parcel_FID field and the final score field is retained (Figure 5). Based on the scoring criteria (Appendix A), a value is assigned to the score field based on percent overlap thresholds (Figure 6).

ClippedBow_AgSoils_FinalTabulateIntersect_Diss

OBJECTID *	Parcel_FID	FREQUENCY	SUM_PERCENTAGE	Bow_AgSoils_Final_Score
1	1	1	100	3
2	2	1	92.811527	3
3	4	1	50.917889	3
4	5	1	100	3
5	6	1	100	3
6	7	1	100	3
7	8	1	64.453107	3
8	9	1	2.280737	1
9	10	1	34.903069	2
10	12	1	54.078737	3

Figure 5. Final score table for the agricultural soils per parcel.

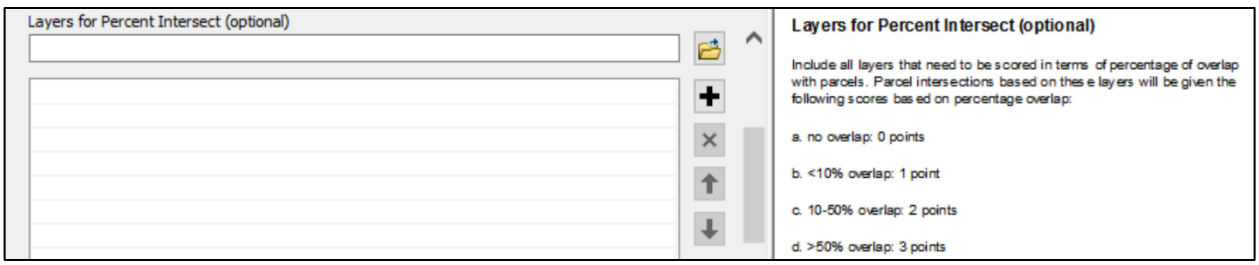

Figure 6. Dialogue box and help menu for the Layers for Percent Intersect parameter.

ii. Rare Natural Communities

Natural communities are assemblages of plants and animals found in particular physical environments that recur over a landscape. New Hampshire contains two to three hundred natural community types, encompassing habitats such as alpine meadows, riverbanks, forests, tidal marshes, ponds, and cliffs. The Natural Heritage Bureau gives conservation priority to "exemplary" or "rare" natural communities. Exemplary natural communities are identified as areas not heavily impacted by humans. These areas have greater potential to contain or achieve natural ecosystem dynamics, characteristic of original community types (before human colonization) [13].

Within the co-occurrence scripts, the natural community feature class is optional along with the Natural Community Field. This field must contain the names of the natural communities present (their types). Once the natural community feature class is provided, a drop-down menu allows the user to select the field specifying the names (e.g. SName). When the tool is run, the natural community data is clipped the study area boundary file. An identifier field is added to the attribute table of the natural community dataset named *Natural Community dataset name* + "_Presence". The Presence field is assigned a value of 1. The table is then intersected with the parcels layer using the Intersect tool.

Clippedexemplary_nat_comm_NHBIntersect

FID *	Shape *	FID_Clippedexemplary_nat_comm_NHB	OBJECTID	EO_ID	SHAPE_ID	SName	Clippedexemplary_nat_comm_NHB_Presence	Parcel_FID
23	Polygon	2	13	7181	40418	Drainage marsh - shrub swamp system	1	3259
34	Polygon	2	13	7181	40418	Drainage marsh - shrub swamp system	1	3259
38	Polygon	2	13	7181	40418	Drainage marsh - shrub swamp system	1	3259
36	Polygon	2	13	7181	40418	Drainage marsh - shrub swamp system	1	3258
32	Polygon	2	13	7181	40418	Drainage marsh - shrub swamp system	1	3249
2	Polygon	1	12	5874	26935	Sand plain basin marsh system	1	2594
1	Polygon	1	12	5874	26935	Sand plain basin marsh system	1	2298

Figure 7. Intersect table showing natural communities per parcel.

After the intersection, the Pivot Table tool is used to manipulate the natural community types and pivot them to become fields rather than records. The Presence score for each parcel per community type is shown in the resulting table (Figure 8).

PivotExemplaryNatComm

FID *	Parcel_FID	Drainage marsh - shrub swamp system	Poor level fen/bog system	Sand plain basin marsh system
1	181	1	0	0
2	244	1	0	0
3	244	1	0	0
4	244	1	1	0
5	381	1	0	0
6	381	1	0	0
7	381	1	1	0
8	429	1	0	0
9	487	1	0	0
10	519	0	1	0
11	534	0	1	0
12	537	1	0	0

Figure 8. Pivot table results, showing presence/absence of each community type per parcel.

As a result of the pivot table, the Parcel_FID are now records in the table, however, there are duplicate values. The Summary Statistics tool allows the table to be "dissolved" by Parcel_FID and statistics computed for the table fields. The co-occurrence script loops through the numeric fields in the pivot table, excluding the Parcel_FID and computes the maximum value for each unique Parcel_FID.

PivotExemplaryNatComm_Statis1

FID *	Parcel_FID	FREQUENCY	MAX_Drainage_marsh___shrub_swamp_system	MAX_Poor_level_fen_bog_system	MAX_Sand_plain_basin_marsh_system
1	181	1	1	0	0
2	244	3	1	1	0
3	381	3	1	1	0
4	429	1	1	0	0
5	487	1	1	0	0
6	519	1	0	1	0
7	534	1	0	1	0
8	537	1	1	0	0

Figure 9. Summary statistics for the natural community pivot table used to "dissolve" Parcel_FID.

The Add Field tool is then used to create a final score field (Figure 10) that evaluates how many unique natural communities are present per parcel. The script searches for the substring

"MAX" contained in the summary statistic table fields and sums these fields. An update cursor then assigns the final scores per parcel based on certain thresholds (Figure 11).

MAX_Drainage_marsh___shrub_swamp_system	MAX_Poor_level_fen_bog_system	MAX_Sand_plain_basin_marsh_system	exemplary_nat_comm_NHB_Score
1	0	0	1
1	1	0	2
1	1	0	2
1	0	0	1
1	0	0	1
0	1	0	1
0	1	0	1
1	0	0	1
1	0	0	1
1	0	0	1

Figure 10. Final score table for the exemplary natural communities per parcel.

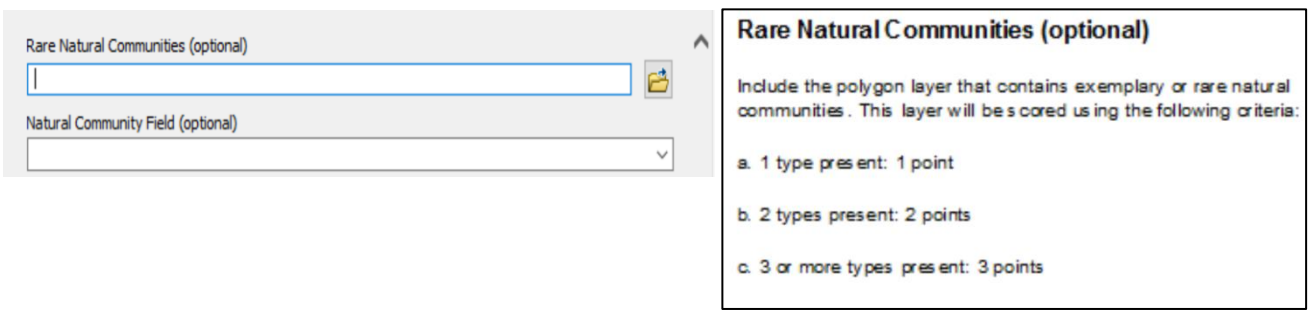

Figure 11. Dialogue box and help menu for the Rare Natural Communities parameter.

iii. Wildlife Action Plan (WAP) Tiers

In 2015, New Hampshire updated its Wildlife Action Plan and in this report GIS analyses were conducted that provide maps of highest ranked habitats by ecological condition for the State. These habitats are grouped into tiers based on their ecological condition in relation to the surrounding landscape. Tiers 1, 2, and 3 refer to overall highest ranked habitat in New Hampshire, highest ranked habitat in the biological region, and supporting landscapes, respectively [11].

In the co-occurrence analysis tool, the WAP Tiers dataset is optional along with the field that contains the WAP tier value (1, 2, or 3). The tier field is available as a dropdown selection once the WAP Tier dataset is input. The WAP Tier dataset is clipped to the study area boundary file and a presence field is created (Section *ii. Rare Natural Communities*). A search cursor then assigns values to the presence field for each WAP tier value. Tier values of 1 are given a value of 1 in the presence field, tier values of 2 are given a value of 2, and tier values of 3 are given a value of 6. This assignment scheme creates unique combinations of tier values per parcel, and will therefore not provide the same sum when added (e.g. 3+0=3 and 2+1=3). Each unique sum combination evaluates to presence of certain tier values within the parcel. For example, a parcel with a sum of 3 contains both tier 1 and tier 2 (1+2=3).

Following the update cursor, the script follows the same steps as the Rare Natural Communities analysis. The data is intersected with the parcels, a pivot table is generated, summary statistics collapse the Parcel_FID and a sum is generated in a new field (Figure 12). Another update cursor takes the unique sums of this presence field and assigns a final score based on the score criteria provided (Figure 13).

ClippedWAPTIERS_PIntersectPivot_Diss

OBJECTID *	Parcel_FID	FREQUENCY	MAX_WAPTIER0	MAX_WAPTIER1	MAX_WAPTIER2	MAX_WAPTIER3	WAPTIERS_P_Score
1	1	2	0	1	0	0	3
2	2	3	0	1	0	0	3
3	3	1	0	0	0	6	1
4	4	1	0	0	0	6	1
5	5	3	0	0	0	6	1
6	6	1	0	0	0	6	1
7	7	1	0	0	0	6	1
8	8	21	0	0	0	6	1
9	9	15	0	0	0	6	1
10	10	12	0	0	2	6	2

Figure 12. Final score table for WAP Tiers per parcel.

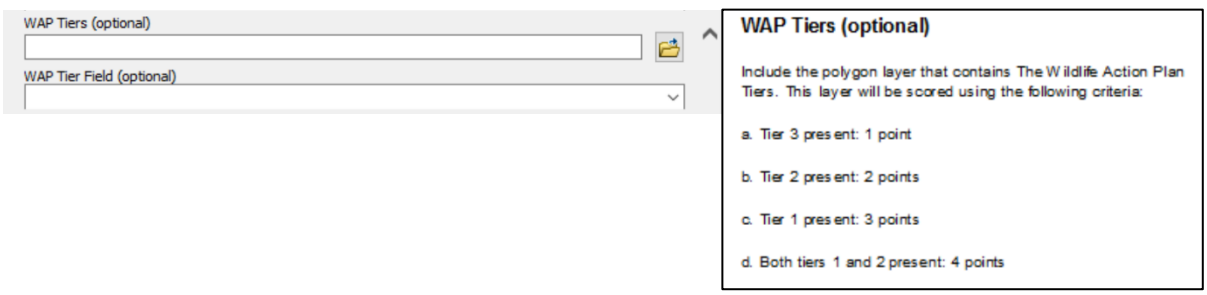

Figure 13. Dialogue box and help menu for the WAP Tiers parameter.

iv. Merrimack Conservation Plan

The Merrimack Conservation Plan, facilitated by the Society for the Protection of NH Forests, was a collaborative conservation planning project conducted in 2014. This report sought to identify CFA in the Merrimack River watershed and Merrimack Valley that contain critical terrestrial and aquatic habitats that contribute to the larger Gulf of Maine ecosystem. Through GIS analysis, 887 individual CFA were identified within the Merrimack Watershed. The Merrimack CFA dataset contains tier values similar to the WAP Tiers, identified through the CFA analysis. The watershed is divided into high, higher, and highest-ranking areas based on a co-occurrence analysis conducted utilizing many environmental datasets [14].

In the co-occurrence analysis tool, the Merrimack Conservation Plan dataset is optional along with the field that contains the tier value (1, 2, or 3). The tier field is available as a dropdown selection once the Merrimack Plan dataset is input. The Merrimack Plan dataset is clipped to the study area boundary file and follows the same analysis steps as the previous section (Section *iii. WAP Tiers).* The Merrimack Plan tiers are evaluated with the scores provided (Figure 14).

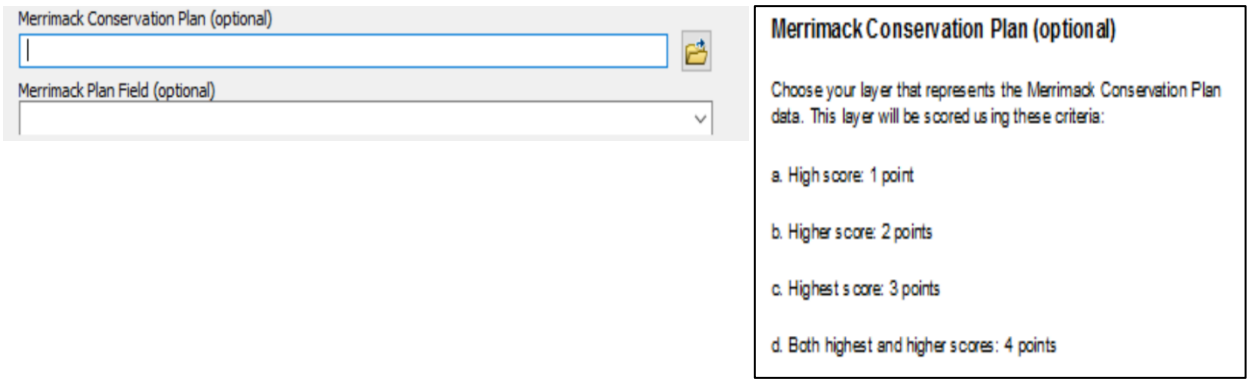

Figure 14. Dialogue box and help menu for the Merrimack Conservation Plan parameter.

v. Unfragmented Lands

Unfragmented lands also known as "unfragmented open space blocks" refer to undeveloped sections of the landscape that are not intersected by roads. As a result, these lands may contain significant wildlife habitats and be areas suitable for outdoor recreation, protection of water resources, etc. Within New Hampshire, blocks vary in size (1 to over 1,000 acres) according to their location or region. Southeastern New Hampshire has a high level of development whereas northern New Hampshire has significantly less development and there are large timberland areas, significant unfragmented blocks are much larger, in the thousands of acres. Unfragmented lands are determined by conducting a GIS analysis whereby NH DOT roads are buffered 500 feet and the buffer is then erased from a polygon file or parcels dataset of the region [1].

In the co-occurrence analysis tool, the Unfragmented Lands dataset analysis is optional. The Unfragmented Lands dataset is clipped to the study area boundary file and a new area field for the unfragmented blocks is calculated within the original data table. The unfragmented lands are intersected with the parcels and dissolved by Parcel FID. Then, a new field is added to the table for the final score per parcel. A search cursor updates the Area field and assigns it a score based on how large the associated unfragmented land block is (Figure 15). Parcels within a larger unfragmented block are assigned a higher score.

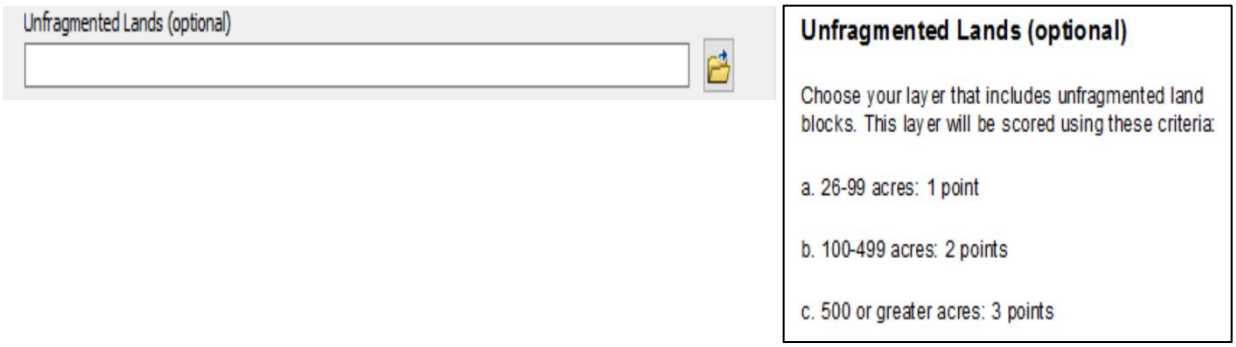

Figure 15. Dialogue box and help menu for the Unfragmented Lands parameter.

vi. Conserved Lands

New Hampshire's Conservation and Public Lands dataset contains information about the conservation status of tax parcels throughout the state. Parcels in the dataset are broken up into categories based on their status (e.g. conservation easement, fee ownership, etc.). Conservation lands are important to this model because lands immediately adjacent to already conserved lands have are designated as high conservation value. This is based on the principle that conservation lands hold valuable resources for wildlife and enlarging existing conserved blocks helps with habitat connectivity. In wildlife reserve design this is termed a "single-large" approach. That is to say, with increasing habitat patch size, the survival chance of a local population is expected to increase [15].

In the co-occurrence analysis tool, a Conservation Lands dataset can be input. The Conserved Lands dataset is clipped to the study are boundary file and The Generate Near Table tool is used to calculate the distance from each parcel to the nearest conservation land.

Bow_Parcels_GenerateNearTabl

OBJECTID *	IN_FID	NEAR_FID	NEAR_DIST
1	1	18	439.256393
2	2	18	19.115212
3	3	18	2738.798216
4	4	18	2806.467521
5	5	18	3419.552332

Figure 16. Near table showing the distance from each parcel to the nearest conservation land.

A final score field is created and the distances are updated via a cursor to yield the final score based on distance thresholds (Figure 17). This table is then joined back to the parcels dataset.

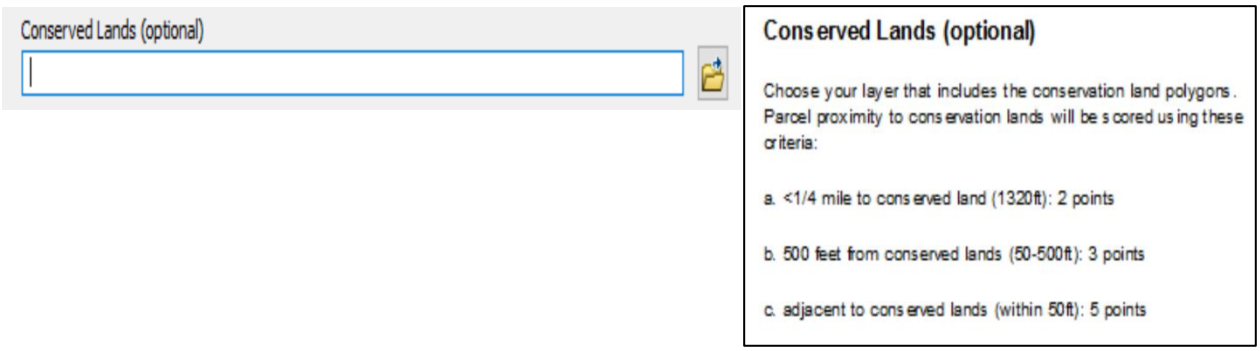

Figure 17. Dialogue box and help menu for the Conserved Lands parameter

vii. Rare Species

Rare species data from the Natural Heritage Bureau was provided by Moosewood Ecological for the Bow NRI. This data provides coarse polygonal data of rare species occurrences. In the co-occurrence analysis tool, the Rare Species dataset is optional along with a field that specifies the names of the rare species present in each polygon record. The Rare Species dataset is handled the same as section *ii. Rare Natural Communities*. Unique species occurrences are aggregated per parcel and a score assigned based on the number of species present (Figure 18).

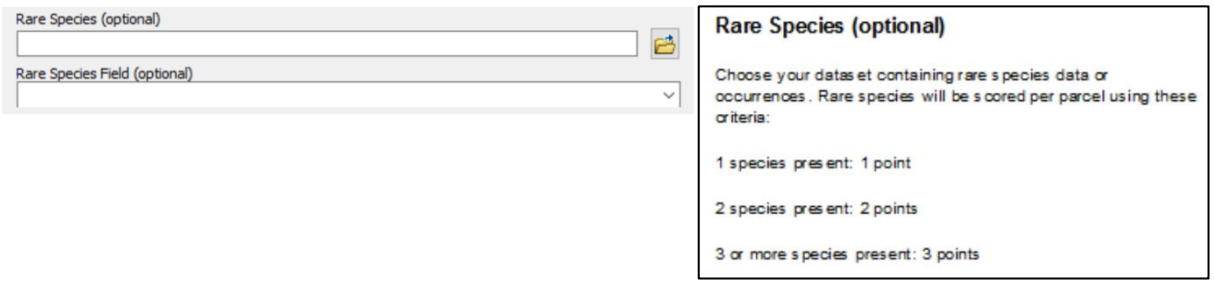

Figure 18. Dialogue box and help menu for the Rare Species parameter.

viii. Area

The last criteria that is added is an evaluation of parcel area. An area field (in acres) is added to the parcels dataset and an update cursor evaluates the area and assigns a score based on the thresholds provided (Appendix A). This analysis does not appear in the tool GUI but is appended as a score to the parcels dataset.

ix. Model Output

The script evaluates all the above factors and sums all of the final score fields that have been joined to the original parcels dataset. A new field is created that stores this sum named AgForestFinal_Ranking (Figure 19). The parcels layer is exported to a new feature class in the Output Directory and is named "AgForestResourcesCoOccurrence".

AgForestResources_CoOccurrence

MerrimackPlan_Clip_Score	Unfrag_Score	ConservedLands_Score	rare_species_NHB_Score	ParcelArea_Score	AgForestFinal_Ranking
2	0	3	0	1	18
2	1	5	0	1	20
1	0	0	0	1	6
1	0	0	0	1	12
1	0	0	0	1	12
1	0	0	1	1	13
1	0	0	1	1	13
1	0	0	1	1	13
1	0	0	0	1	10
2	1	5	0	1	18
1	1	5	0	1	12
1	1	3	0	1	16
2	2	0	1	1	14
2	2	0	1	1	14
2	2	0	0	1	6

Figure 19. Final analysis output for the Agricultural and Forest Resources Model (AgForestFinal_Ranking). This table shows a subset of the score fields.

2.3.3 Water Resources Co-occurrence Model

The Water Resources Co-occurrence is a parcel-based GIS assessment model that provides a sense of the relative water resources value for each parcel in the study area. Similar to the *Agricultural and Forest Resources Co-occurrence Model*, this tool aids towns and conservation commissions by outlining desired areas for land conservation, and spurring engagement with property owners. The assessment for each parcel includes many attributes: wetlands, hydric soils, ponds, streams, aquifers, favorable gravel well analysis, wellhead protection areas, NH designated corridor of the Merrimack River, regional conservation plans, size of the property, fragmentation, proximity to conserved lands, and rare species. The following parameters for this model have been described in the previous model and the analysis steps are the same: WAP Tiers, Merrimack Conservation Plan, Unfragmented Lands, Conservation Lands, and Area.

i. Layers for Percent Intersect

For details on the analysis associated with this parameter refer to *Agricultural and Forest Resources Co-occurrence Model (*Section *i.).* For this specific model, Layers for Percent Intersect may include wetlands, hydric soils, and other datasets as specified by the user.

ii. Waterbodies

Waterbodies refer to New Hampshire ponds and lakes. This data comes from the New Hampshire Hydrography Dataset (NHHD) and maps displaying this data are recommended for NRI in New Hampshire [1]. In the co-occurrence analysis tool, the Waterbodies dataset is clipped to the study area boundary file, an area field is added to Waterbodies in acres, and an update cursor then assigns a score to Waterbodies based on size (Figure 20). The Waterbodies dataset is then intersected with the parcels dataset and the Area field (now containing scored values) is renamed as a score field, similar to previous parameters.

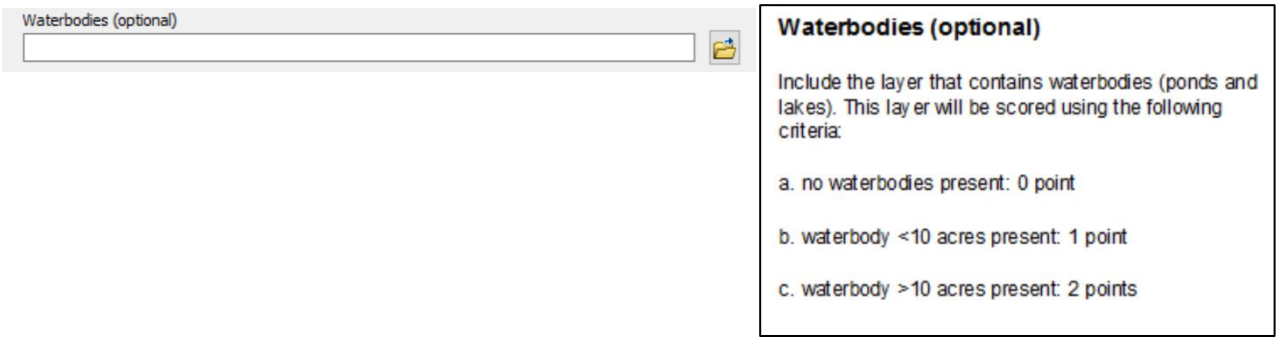

Figure 20. Dialogue and help menu for the Waterbodies parameter.

iii. Watercourses

Watercourses refer to New Hampshire streams (flowlines). This data likewise comes from the NHHD and this data is recommended for NRI in New Hampshire [1]. In the co-occurrence analysis tool, the Watercourses dataset is optional along with the field that specifies the stream order. The Watercourses dataset is clipped to the study area boundary file. The script adds an identifier field and assigns a value to this field based on stream order and the scoring criteria (Figure 21). Any stream above order 4 (Strahler classification) is assigned a value of 4. Orders below 4 retain their original values (e.g. order 1=1). The streams are then intersected with the parcels and summary statistics calculate the max stream order present in each parcel. The presence field is renamed as the final score field.

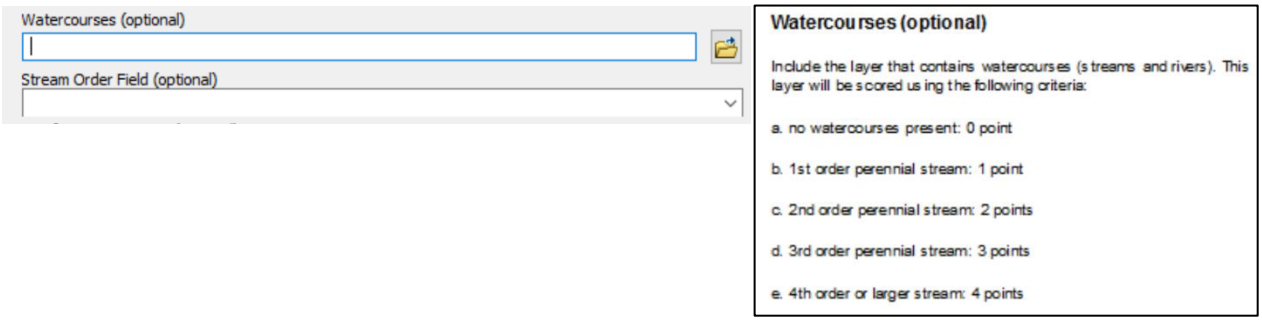

Figure 21. Dialogue and help menu for the Watercourses parameter.

iv. Aquifer Transmissivity

More than 50% of New Hampshire's population depends on groundwater sources for drinking water. Groundwater is found below the land surface in the pore spaces between grains of sands, silts and clays or in the fracture planes of bedrock. Aquifers are saturated zones in sediments, such as sand and gravel, and in fractured rock formations that receive, store and transmit water to wells and springs [1]. Aquifers serve as an important source of groundwater for commercial, industrial, domestic, and public-water supplies in the State [16].

Stratified drift aquifers are unique as they are underlain by unconsolidated glacial sediments that began to form during late stages of the Great Ice Age, about 14,000 years ago. The southernmost extent of the most recent continental glacier had retreated from positions on Long Island, New York, to positions in New Hampshire during this time in a step-wise fashion. Glacial lakes that ponded in front of the melting ice margin played an important role in the formation o£ stratified drift aquifers in New Hampshire, serving as natural sediment traps. Due to the vast extent of the last glacial maximum, these aquifers are ubiquitous in the northeastern United States. Approximately 14% of New Hampshire is underlain by these aquifers. The transmissivity of these groundwater resources, usually presented in square feet transmitted per day (ft^2/d), refers to the ability of the aquifer to store and transport water and is determined by the porosity of the substrate [16].

In the co-occurrence analysis tool, the Aquifer Transmissivity (Stratified Drift Aquifer) dataset is optional along with the field that specifies the aquifer transmissivity class. The transmissivity classes are as follows: 1= <2000ft^2/d, 2=2000-4000ft^2/d, and 3= >4000ft^2/d. The Aquifer Transmissivity dataset is clipped to the study area boundary file and intersected with the parcels dataset. The transmissivity field is renamed to “Transmissivity_Score”, and summary statistics are used to dissolve the table by Parcel_FID and calculate the maximum transmissivity value per parcel. For instance, if a parcel contains an aquifer with a class transmissivity of 1 and 2, the parcel will be given a score of 2. The scores allocated to the parcel is based on scoring criteria (Figure 22).

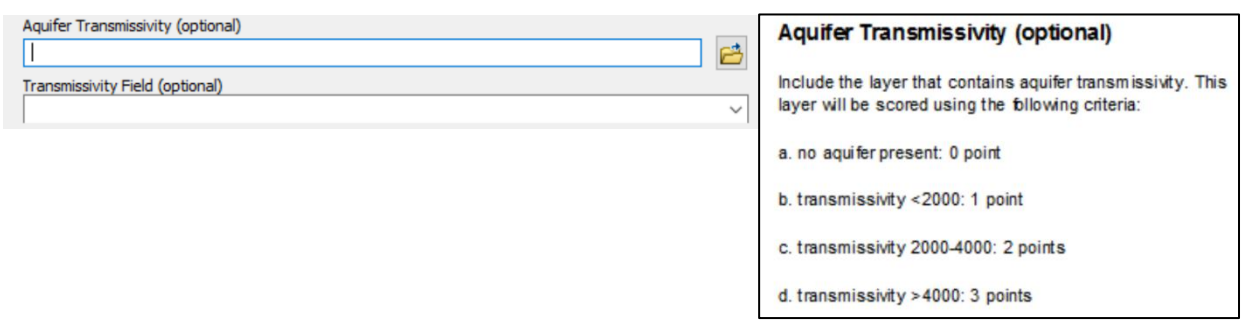

Figure 22. Dialogue and help menu for the Aquifer Transmissivity parameter.

v. Favorable Gravel Well Analysis

Favorable gravel well analysis refers to a GIS technique for analyzing groundwater resources developed by the New Hampshire Department of Environmental Services. This technique aims to help water suppliers and community planners take into account water quality and quantity constraints when analyzing stratified drift aquifer maps to review potential sites for new community wells [1]. These wells are mapped according to their ability to produce water in gallons per minute (gpm).

In the co-occurrence analysis tool, the Favorable Gravel Well Analysis is split into two parameters. One parameter is a dataset containing gravel wells that produce water at 75gpm and the other is wells producing at 150gpm. The gravel well datasets are both clipped to the study area boundary file. A new field is added that gives gravel wells (75gpm) a value of 1 and gravel wells (150gpm) a value of 2. The two datasets are then merged using the Merge tool and these fields are retained. The resulting dataset is intersected with the parcels dataset and dissolved using the Dissolve tool and the max value for gravel well presence is calculated. A parcel containing both gravel well types (75 and 150gpm) would be given a score of 2 (the max score present within the parcel). Parcels can have a score of 0, 1, or 2 (Figure 23). This score field is renamed to "GravelWell_Score" in the final co-occurrence output.

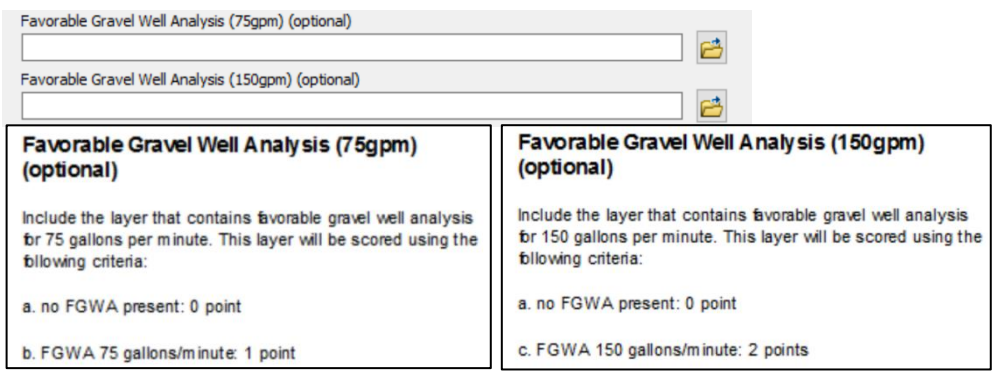

Figure 23. Dialogue and help menu for the Favorable Gravel Well Analysis parameter.

vi. Wellhead Protection Areas

Groundwater is recharged (replenished) when rain and snow melt infiltrate down into the saturated zones of soil. Groundwater quality in NH is normally good, however, groundwater is vulnerable to contamination from a variety of sources such as: chemical spills, leakage from landfills and underground storage tanks, septic systems, road salt, etc. Wellhead Protection Areas (WHPA) are the areas surrounding a public water supply. New Hampshire's wellhead protection program enables municipalities, water suppliers and local entities to delineate protection areas, identify contamination sources and actively manage land use in the WHPA. Increased impervious surfaces reduce infiltration to groundwater and increase surface runoff. Highly

developed areas and impervious services can have a significant effect on recharge to groundwater supplies [1].

In the co-occurrence analysis tool, Wellhead Protection Areas are an optional parameter. The dataset is clipped to the study area boundary file and intersected with the parcels dataset. The intersection table is dissolved by Parcel_FID using the Dissolve tool. A field is added that contains a value of 1 for those parcels containing a WHPA and this table is joined to the parcels dataset (Figure 24). Parcels not containing a WHPA are assigned a value of NULL, which are converted to zeros at the end of the co-occurrence script.

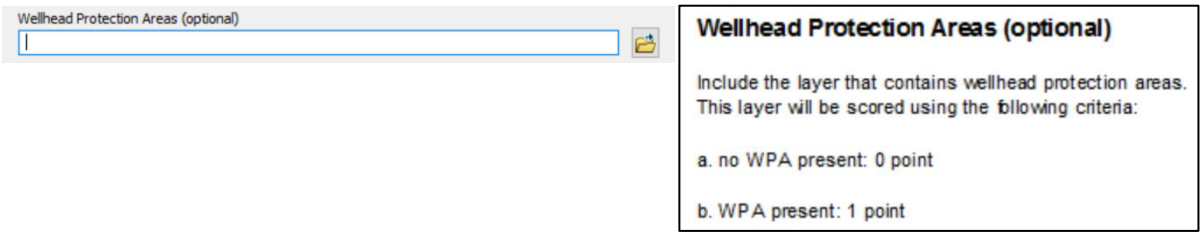

Figure 24. Dialogue and help menu for the Wellhead Protection Areas parameter.

vii. Rare species

For this model, only rare species that heavily utilize water resources are included in the input dataset. Tool users will need to manually create a subset of aquatic rare species from the NHB dataset.

viii. River Corridor

The Merrimack Conservation Plan also designated significant riparian corridors. The Merrimack River Corridor dataset is a polygon buffer of .25 miles from either side of the Merrimack River. The River Corridor parameter can be any significant riparian corridor and is an optional input. The River Corridor dataset is clipped to the study area boundary file. A presence field is added to the table (value of 1 for parcels intersecting the corridor). The River Corridor dataset is then intersected with the parcels dataset and the presence field is renamed "RiverCorridor_Score". Any parcel not containing a riparian corridor is given a value of zero, parcels having a corridor retain a value of 1 (Figure 25).

Figure 25. Dialogue and help menu for the River Corridor parameter.

ix. Model Output

The script evaluates all the above factors and sums all of the final score fields that have been joined to the original parcels dataset. A new field is created that stores this sum named "WaterResourcesFinal_Ranking". The parcels layer is exported to a new feature class in the Output Directory and is named "WaterResources_CoOccurrence". Note users should remove features not containing water from the water co-occurrence manually before entering the result into the CFA tool if they so desire. To do so, simply sum all the fields containing water features: hydric soils, wetlands, waterbodies, watercourses, wellhead protection areas, and aquifer transmissivity. If the sum equates to zero, remove these records from the table.

2.3.4 Ecological Resources Co-occurrence Model

The Ecological Resources Co-occurrence is a parcel-based GIS assessment model that provides a sense of the relative ecological resources value for each parcel in the study area. This assessment highlights wildlife resources and evaluates many attributes such as: wildlife habitats, declining/vulnerable habitats, rare/unique forest communities, regional conservation plans, size of the property, fragmentation, proximity to conserved lands, and rare species. The following parameters for this model have been described in the previous models and the analysis steps are the same: Layers for Percent Intersect, Waterbodies, Watercourses, Rare Natural Communities, WAP Tiers, Merrimack Conservation Plan, Unfragmented Lands, Conserved Lands, Rare Species, and Area.

i. Wildlife Habitats

New Hampshire contains a wide variety of habitats, from coastal islands to alpine tundra [1] and these are documented in The Wildlife Action Plan data from NH GRANIT. For the purposes of these co-occurrence tools, developed classes of this dataset along with open water, peatlands, and grasslands were excluded. Parcels are evaluated based on the number of unique wildlife habitats present.

In the co-occurrence analysis tool, the Wildlife Habitats dataset is evaluated along with a field containing the habitat names (similar to *Rare Natural Communities*). The tool clips the dataset to the study area boundary, populates the table with a field containing values of 1. The Wildlife Habitats are then intersected with the parcels layer, a pivot table makes each unique habitat type its own field, and then a final score based on the scoring criteria (Figure 26) is generated by summing all the types present within a parcel.

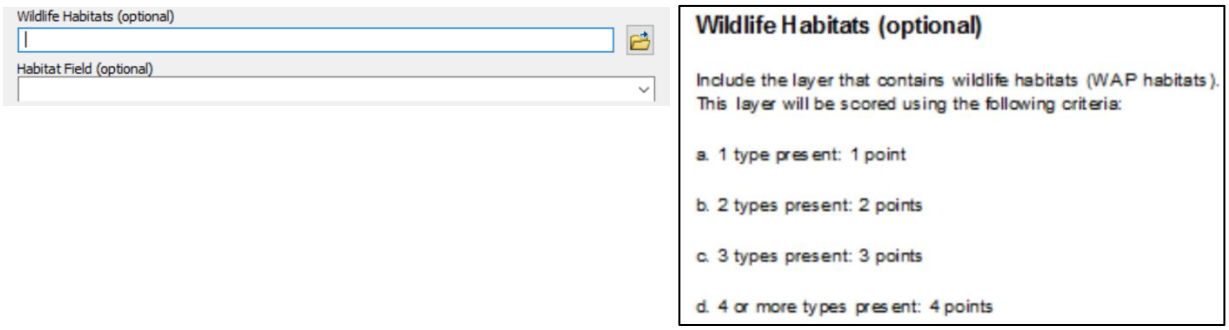

Figure 26. Dialogue and help menu for the Wildlife Habitats parameter.

ii. Rare Habitats

Some habitats in New Hampshire such as shrublands, peatlands, grasslands, and vernal pools, are considered vulnerable, declining, or regionally rare. For this analysis, shrublands were obtained from transmission line data. Peatlands and grasslands were extracted from the Wildlife Action Plan data. Vernal pools were obtained as a separate dataset provided by Moosewood Ecological. All of these datasets were merged into one input feature class. In order to merge the transmission line and vernal pool data, these datasets were buffered by 1 foot to transform them into polygon features. In the co-occurrence analysis tool, the Rare Habitats dataset along with a field containing the unique habitat names is analyzed in the same fashion at the Wildlife Habitats data (Section *iv. Wildlife Habitats*). Each parcel is assigned a score in terms of the number of habitats present (Figure 27).

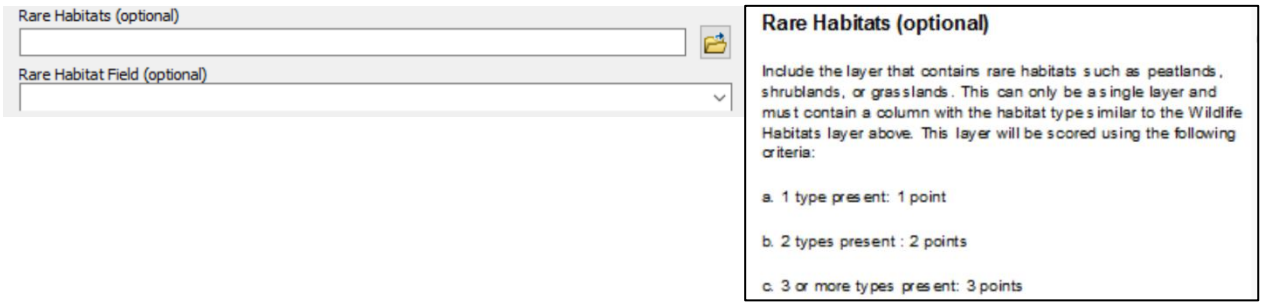

Figure 27. Dialogue and help menu for the Rare Habitats parameter.

iii. Model Output

The script evaluates all the above factors and sums all of the final score fields that have been joined to the original parcels dataset. A new field is created that stores this sum named "EcoFinal_Ranking". The parcels layer is then exported to a new feature class, named "EcologicalResources_CoOccurrence", in the Output Directory.

2.3.5 Weighted-Combined Co-occurrence Model

The Weighted-Combined Co-occurrence Model is a summation of the previous three co-occurrence model outputs: Agricultural Forest Resources, Water Resources, and Ecological Resources. Within the toolbox, the script tool is referred to as "Conservation Focal Areas". This tool is designed to help stakeholders identify conservation priorities by overlapping the three co-occurrence models. Users may elect to assign weights to each input co-occurrence model (any positive numerical value), where the weight determines the "importance" of a particular co-occurrence model in the output. The script spatially unions all three of the co-occurrence models and deletes all fields from the attribute table except the final score fields. Fields are added for the weighting factors specified by the user. A field calculator expression then multiplies each co-occurrence score field by its weighting factor and sums the resulting scores:

$$\frac{\begin{gathered}(\text{AgForestFinal_Ranking} \times \text{AgForest_Weight}) \\ + \\ (\text{WaterResourcesFinal_Ranking} \times \text{Water_Weight}) \\ + \\ (\text{EcoFinal_Ranking} \times \text{Ecological_Weight})\end{gathered}}{\text{ConservationFocal_Score}}$$

The tool creates a new feature class "ConservationFocalAreas" within the specified Output Directory.

2.3.6 Model Validation

In order to assess the accuracy of the automated co-occurrence analyses, the output feature classes of the co-occurrence scripts were compared to manually-conducted co-occurrence analyses completed in the December of 2019 for the town of Bow, NH. In order to directly compare the two methods, the script output was spatially joined with its companion manually-calculated dataset and all attributes were joined per parcel. An integer field was added to the resulting join table and a field calculator expression was used to calculate the absolute value of the difference between the two final score parameters. This difference field gives the absolute difference in terms of the final score value between the two analysis methods. Percent difference was calculated in another field and is a better reflection of the difference between the automated and manual-scored values (Figure 28). For the manual Water Resources Co-occurrence Model, all features were removed that did not contain water features in the original analysis. However, for validation purposes, all parcel records were retained regardless of whether they contained water features in order to evaluate if all score parameters were evaluated correctly in relation to the automated method.

AgForest_Validate

Parcel_FID	AgForestFinal_Ranking	Final_Scor	Difference	PercentDiff
1	18	16	2	11.764706
2	20	20	0	0
3	6	6	0	0
4	12	12	0	0
5	12	12	0	0
6	13	13	0	0
7	13	13	0	0

Figure 28. Example of a validation feature class for the Agricultural Forest Co-occurrence. AgForestFinal_Ranking is the final score field for the automated Agricultural Forest Resources Co-occurrence output while Final_Scor was the manually-calculated output.

3. Results

3.1 Parcel Score Assessment

For the Agricultural and Forest Resources Co-occurrence, the two score fields varied by an average of .025 points (.186% difference). The Water Resources Co-occurrence and Ecological Resources Co-occurrence varied by 3.67 and 1.73 points, respectively. This equates to 49.4% difference and 31.8% difference (Figure 29). The highest score differences for all models existed where there were edge-match errors associated with the spatial join.

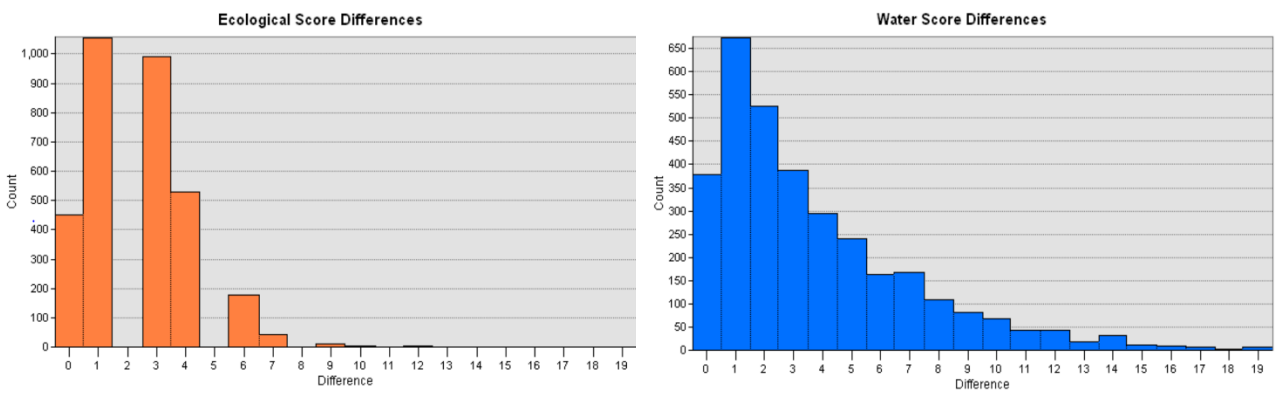

Figure 29. Histograms of the score differentials for the Ecological Resources Co-occurrences and Water Resources Co-occurrences.

For each of the parcel-based assessments, analyses were conducted for all 3,263 parcels within the town of Bow. In all four models, no parcels received a score of zero and the minimum value for any parcel was a score of 1. The automated Agricultural and Forest Resources Model achieved the highest mean score (M = 13.37) of all the individual co-occurrence models and the lowest standard deviation from the mean (SD = 3.98;Table 2).

Table 2. Summary statistics for the four automated parcel-based assessments.

Model	Min	Max	Mean	Standard Deviation
Agricultural Forest Resources	1	29	13.37	3.98
Water Resources	1	31	7.48	4.71
Ecological Resources	1	31	6.45	4.44
Conservation Focal Areas	5	90	26.04	12.80

Parcels obtaining high scores for all three of the individual co-occurrence models exist in areas adjacent to existing conservation lands, areas that are unfragmented, and where parcels are large in area. When all three co-occurrence models are summed to produce the Conservation Focal Areas, parcels with high scores (≥ 2.5 std. dev., mean=26.04) existed in areas adjacent to existing conservation lands where roads and other development are lacking (Appendix E).

3.2 Script Efficiency

Parcel-based co-occurrence analyses can take an experienced GIS technician 20-40 hours (or more) to complete depending on the number, size, and complexity of datasets analyzed for the study. The co-occurrence script toolset contains 1,985 lines of code that can be run in approximately 27 minutes. Accounting for time to input datasets and analyze the outputs, the total completion time for the Bow analysis was approximately 40 minutes using the automated approach (as opposed to 40 hours for the manual method), equating to a 98.33%-time reduction.

Table 3. Co-occurrence script environment summary and run-times for the Bow, NH NRI.

Script	Lines of Code	Run-Time [mm:ss]
Agricultural and Forest Resources	513	07:16
Water Resources	635	10:25
Ecological Resources	736	08:16
Conservation Focal Areas	101	00:32

4. Discussion

The discrepancies between the scores of the manually-conducted versus automated approaches to the co-occurrence models were stark in terms of percent difference. However, when looking at the score differential, these differences only equate to a few points. Point divergence was often due to edge-matching and join errors. The highest percent difference values for the three co-occurrences existed in parcels that had been cut-off near the edge of the Bow town boundary. When these parcels were spatially joined, incorrect scores were assigned from the parcel being joined. Ideally, an original ID field would have been created for the parcels layer before any co-occurrence analysis was conducted so that the manual result could be joined

to the automated result using the ID as a common field. This would allow for a direct comparison of the final score values and mitigate any spatial join errors.

Second, point divergence occurred because inconsistencies were identified in the manual analysis methods associated with improper transformation of the input datasets (pre-processing). For example, improper creation of the forest land cover layer deducted points from many parcels in the manually-generated Agricultural and Forest Resources Co-occurrence, adding to the score differential. For the ecological co-occurrence, peatlands were included in the habitat analysis for WAP Habitats and Rare Habitats, potentially inflating the scores in parcels containing peatlands. Differences arose due to alternate analysis methods. Using Select by Attribute for the manual methods versus the Intersect tool for the automated analyses lead to inconsistencies in how some parcels were coded in terms of their value for various parameters (e.g. presence of streams). When inconsistencies associated with pre-processing error of the data are accounted for in the manual method, score divergence for the analysis methods is minimalized.

Redundant parameters in the co-occurrence models created a stark spatial pattern in term of final CFA scores. High-scoring areas in the co-occurrence analysis outputs were in areas adjacent to existing conservation lands, within unfragmented land blocks, and in parcels with large areas; all of these attributes are included within each model. If combining the three co-occurrence analyses in the future, redundancy in model attributes should be reduced to alleviate inflated scores. Parcels adjacent to conservation lands receive a score of 5 (the highest score of any criteria possible), leading to a score inflation of 15 points if all three co-occurrences are included in the CFA analysis, as each model scores conservation lands separately.

5. Conclusions

Multiple ecological datasets can be transformed, evaluated and overlaid using professional knowledge in a GIS to inform land management decisions. Using a co-occurrence analysis approach by parcel allows for the delineation of CFA. Cultural, historical, and ecological datasets can be intersected with parcels within a town or region to calculate a percent overlap or presence/absence of significant features on the landscape. Once the percent overlap is calculated, percent thresholds or the number of unique features can be transformed to a preference scale in terms of conservation value, determined by conservation professionals. Scoring-based approaches are easily understood by end-users, especially those working in conservation planning.

Conducting co-occurrence analyses manually often leads to conflated user-error associated with extracting individual attributes and intersecting them with the parcels layer, using select by attribute, and manually-coding values in the attribute tables. By using Python to automate co-occurrence analyses in ArcGIS, technicians can significantly improve accuracy and time-efficiency. The manual methods take an experienced GIS technician 20-40 hours to complete, whereas the automated models can be run in 20-40 minutes (98.33% time reduction). By creating

standalone scripts for ASAL to use for Moosewood Ecological NRI, ASAL staff can complete projects quickly, accurately, and consistently. The co-occurrence scripts and parameters can be adjusted along with the scoring criteria by ASAL technicians, allowing for model flexibility, which can be marketed to clients.

Models outlined in this document have inherent limitations, as they function on polygon feature classes exclusively. Data inputs are aggregated primarily from NH GRANIT and the scripts function using specified scoring criteria. Run-times vary depending on the number of datasets included in the analysis. Model improvements in the future might include: building a parameterized ModelBuilder workflow, improving processing time, automating the results validation, and adding code to symbolize the output parcel layers in terms of conservation value.

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15. Ovaskainen, O. Long-term persistence of species and the SLOSS problem. *Journal of Theoretical Biology* **2002**, *218*, 419–433.
16. Medalie, L.; Moore, R.B. *Ground-water resources in New Hampshire; stratified-drift aquifers*; US Geological Survey; Earth Science Information Center, Open-File Reports …, 1995.

Deliverables

- **Co-occurrence Toolbox:** The "Co-occurrence.tbx" toolbox has been provided to ASAL for use in NRI for Moosewood Ecological, LLC. The toolbox has been uploaded to the ASAL Google Drive. This toolbox contains: Agricultural Forest Resources Co-occurrence, Water Resources Co-occurrence, Ecological Resources Co-occurrence, and Conservation Focal Areas script tools.
- **Scripts:** The Python script files for each of the co-occurrence tools has been uploaded to the ASAL Google Drive.
- **Maps:** 11x17 maps depicting all three automated co-occurrences and CFA (4 maps total) for Bow have been provided to Moosewood Ecological, LLC. Maps were shared in .jpeg and .pdf formats.
- **Bow Geodatabases:** Geodatabases containing all data used in the automated co-occurrence models and their outputs were uploaded to the ASAL Google Drive.

Appendix A: Model Criteria

Parcel-based GIS Ecological Assessment Model
Agriculture and Forest Resources
Bow, NH

This model was designed to assess parcels in a geographical information system (GIS) using the natural resources data that was incorporated into the Bow Natural Resources Inventory.

I. Agricultural Resources (active farmland from Moosewood; ag soils from GRANIT combining 3 types of ag soils together as one data set for this analysis)

Percent Active Farmland

a. no active farmland	0 points
b. <10% active farmland	1 point
c. 10-50% active farmland	2 points
d. >50% active farmland	3 points

Percent Agricultural Soils

a. no agricultural soils	0 points
b. <10% agricultural soils	1 point
c. 10-50% agricultural soils	2 points
d. >50% agricultural soils	3 points

II. Forest Resources (forest cover from land cover data from GRANIT or a better source based on your advice; soils data from GRANIT combining the 3 types together as one data set for this analysis)

Percent Forest Cover (land cover or better data)

a. no forest cover	0 point
b. <10% forest cover	1 point
c. 10-50% forest cover	2 points
d. >50% forest cover	3 points

Percent Forest Soils (combine classes 1A, 1B, and 1C)

a. no forest soils	0 point
b. <10% forest soils	1 point
c. 10-50% forest soils	2 points
d. >50% forest soils	3 points

III. Rare/unique natural communities (data from Moosewood)

a. 1 type present	1 point
b. 2 types present	2 points
c. 3 or more types present	3 points

IV. Conservation Focal Areas (WAP tiers from GRANIT; Merrimack Plan data from Moosewood)

WAP Tiers

a. Tier 3 present	1 point
b. Tier 2 present	2 points
c. Tier 1 present	3 points
d. Both tiers 1 and 2 present	4 points

Merrimack Conservation Plan

a. High score	1 point
b. Higher score	2 points
c. Highest score	3 points
d. Both highest and higher scores	4 points

V. Associated Unfragmented Block Size (data from GRANIT)

a. 26-99 acres	1 point
b. 100-499 acres	2 points
c. 500-999 acres	3 points

VI. Proximity to Conserved Lands (data from Moosewood)

a. <1/4 mile to conserved land	2 points
b. 500 feet from conserved lands	3 points
c. adjacent to conserved lands	5 points

VII. Parcel Size (data from GRANIT)

a. 0-10 acres	1 point
b. 11-50 acres	2 points
c. 51-100 acres	3 points
d. >100 acres	4 points

IX. Rare species (data from Moosewood)

a. 1 species present	1 point
b. 2 species present	2 points
c. 3 or more species present	3 points

Parcel-based GIS Ecological Assessment Model

Water Resources
Bow, NH

This model was designed to assess parcels in a geographical information system (GIS) using the natural resources data that was incorporated into the Bow Natural Resources Inventory. All data is available from GRANIT except as noted. ASAL should have data previously used for the draft NRI maps. Moosewood Ecological has provided the additional data needed as noted below.

***Only rank parcels that contain one or more of the following water resources: wetlands, hydric soils, waterbodies, water courses, aquifers, and wellhead protection areas.**

I. Wetlands (NWI Plus data from GRANIT)

Percent Wetland Cover

a. no wetlands	0 point
b. <10% wetlands	1 point
c. 10-50% wetlands	2 points
d. >50% wetlands	3 points

II. Hydric Soils (combine poorly drained and very poorly drained soils together into one polygon to use as "hydric soils'; data from GRANIT)

a. no hydric soils	0 point
b. <10% hydric soils	1 point
c. 10-50% hydric soils	2 points
d. >50% hydric soils	3 points

III. Waterbodies (ponds/open water from Moosewood)

a. no waterbodies present	0 point
b. waterbody <10 acres present	1 point
c. waterbody >10 acres present	2 points

IV. Watercourses (streams from GRANIT)

a. no watercourses present	0 point
b. 1st order perennial stream	1 point
c. 2nd order perennial stream	2 points
d. 3rd order perennial stream	3 points
e. 4th order or larger stream	4 points

V. Stratified Drift Aquifers (aquifers/transmissivity from GRANIT; FGWA from DES

Transmissivity

a. no aquifer present	0 point
b. transmissivity <2000	1 point
c. transmissivity 2000-4000	2 points
d. transmissivity >4000	3 points

Favorable Gravel Well Analysis (FGWA)

a. no FGWA present	0 point
b. FGWA 75 gallons/minute	1 point
c. FGWA 150 gallons/minute	2 points

VI. Wellhead Protection Areas (WPA; data from DES)

a. no WPA present	0 point
b. WPA present	1 point

VII. Conservation Focal Areas (WAP tiers from GRANIT; Merrimack Plan data from Moosewood)

WAP Tiers

a. Tier 3 present	1 point
b. Tier 2 present	2 points
c. Tier 1 present	3 points
d. Both tiers 1 and 2 present	4 points

Merrimack Conservation Plan

a. High score	1 point
b. Higher score	2 points
c. Highest score	3 points
d. Both highest and higher scores	4 points

VIII. Associated Unfragmented Block Size (data from GRANIT)

a. 26-99 acres	1 point
b. 100-499 acres	2 points
c. 500-999 acres	3 points

IX. Proximity to Conserved Lands (data from Moosewood)

a. <1/4 mile to conserved lands	2 points
b. 500 feet from conserved lands	3 points
c. adjacent to conserved lands	5 points

X. Parcel Size (data from GRANIT)

a. 0-10 acres	1 point
b. 11-50 acres	2 points
c. 51-100 acres	3 points
d. >100 acres	4 points

XI. Rare species (only those associated with surface waters and wetlands; data from Moosewood)

a. 1 species present	1 point
b. 2 species present	2 points
c. 3 or more species present	3 points

XII. NH Designated River Corridor – Merrimack River (data from Moosewood)

a. Not present	0 point
b. Present	1 point

Parcel-based GIS Ecological Assessment Model
Ecological Resources
Bow, NH

This model was designed to assess parcels in a geographical information system (GIS) using the natural resources data that was incorporated into the Bow Natural Resources Inventory.

I. Waterbodies (ponds/open water from Moosewood)

a. no waterbodies present	0 point
b. waterbody <10 acres present	1 point
c. waterbody >10 acres present	2 points

II. Watercourses (streams from GRANIT)

a. no watercourses present	0 point
b. 1st order perennial stream	1 point
c. 2nd order perennial stream	2 points
d. 3rd order perennial stream	3 points
e. 4th order or larger stream	4 points

III. Wildlife Habitats (WAP habitats from GRANIT)

a. 1 type present	1 point
b. 2 types present	2 points
c. 3 types present	3 points
d. 4 or more types present	4 points

IV. Rare/declining/vulnerable habitats (vernal pools, grasslands, shrublands, peatlands data from Moosewood)

a. 1 type present	1 point
b. 2 types present	2 points
c. 3 or more types present	3 points

V. Rare/unique natural communities (data from Moosewood)

a. 1 type present	1 point
b. 2 types present	2 points
c. 3 or more types present	3 points

VI. Conservation Focal Areas (WAP tiers from GRANIT; Merrimack Plan data from Moosewood)

WAP Tiers

a. Tier 3 present	1 point
b. Tier 2 present	2 points
c. Tier 1 present	3 points
d. Both tiers 1 and 2 present	4 points

Merrimack Conservation Plan

a. High score	1 point
b. Higher score	2 points
c. Highest score	3 points
d. Both highest and higher scores	4 points

VII. Associated Unfragmented Block Size (data from GRANIT)

a. 26-99 acres	1 point
b. 100-499 acres	2 points
c. 500-999 acres	3 points

VIII. Proximity to Conserved Lands (data from Moosewood)

a. <1/4 mile to conserved lands	2 points
b. 500 feet from conserved lands	3 points
c. adjacent to conserved lands	5 points

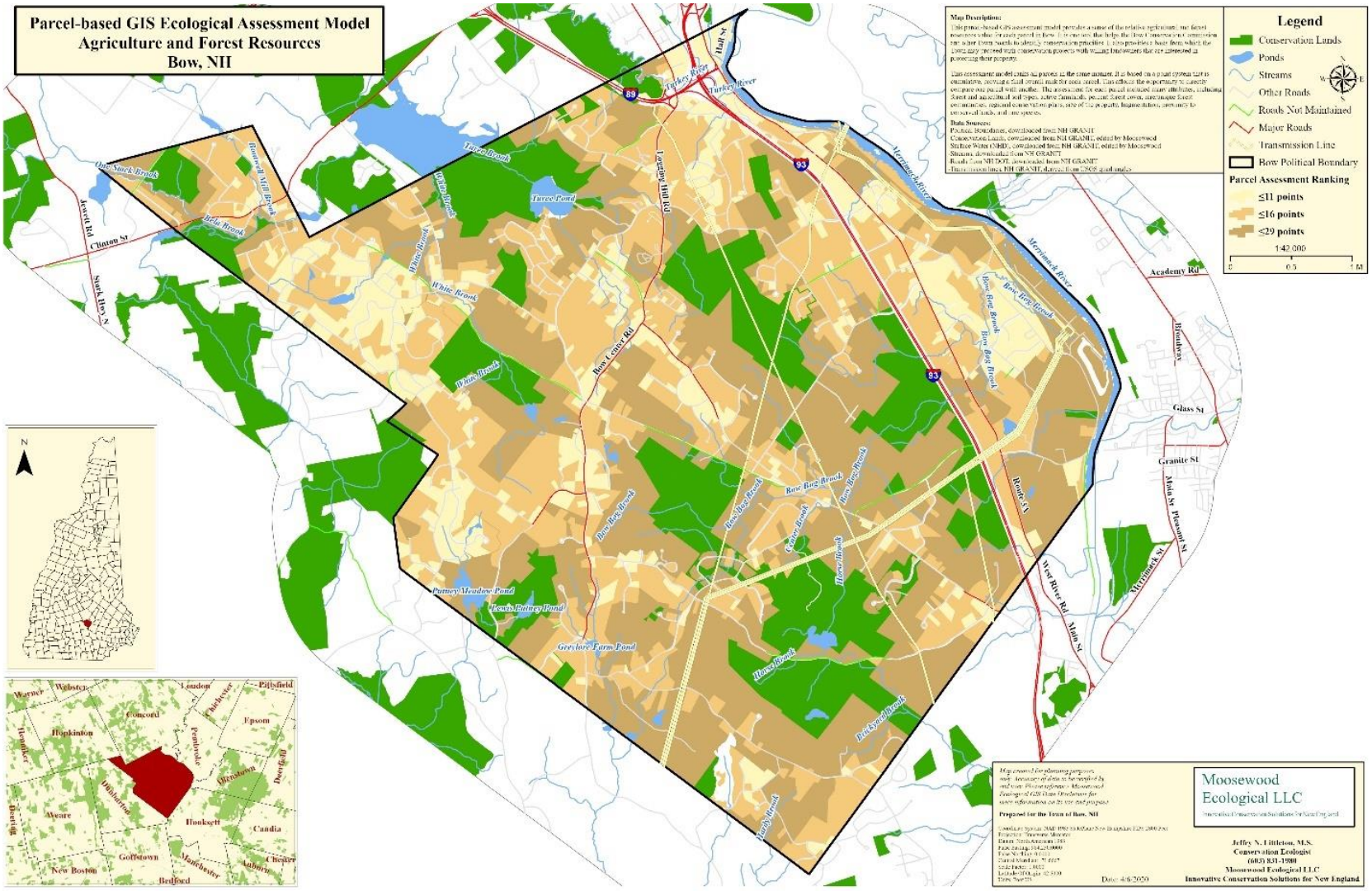
IX. Parcel Size (data from GRANIT)

a. 0-10 acres	1 point
b. 11-50 acres	2 points
c. 51-100 acres	3 points
d. >100 acres	4 points

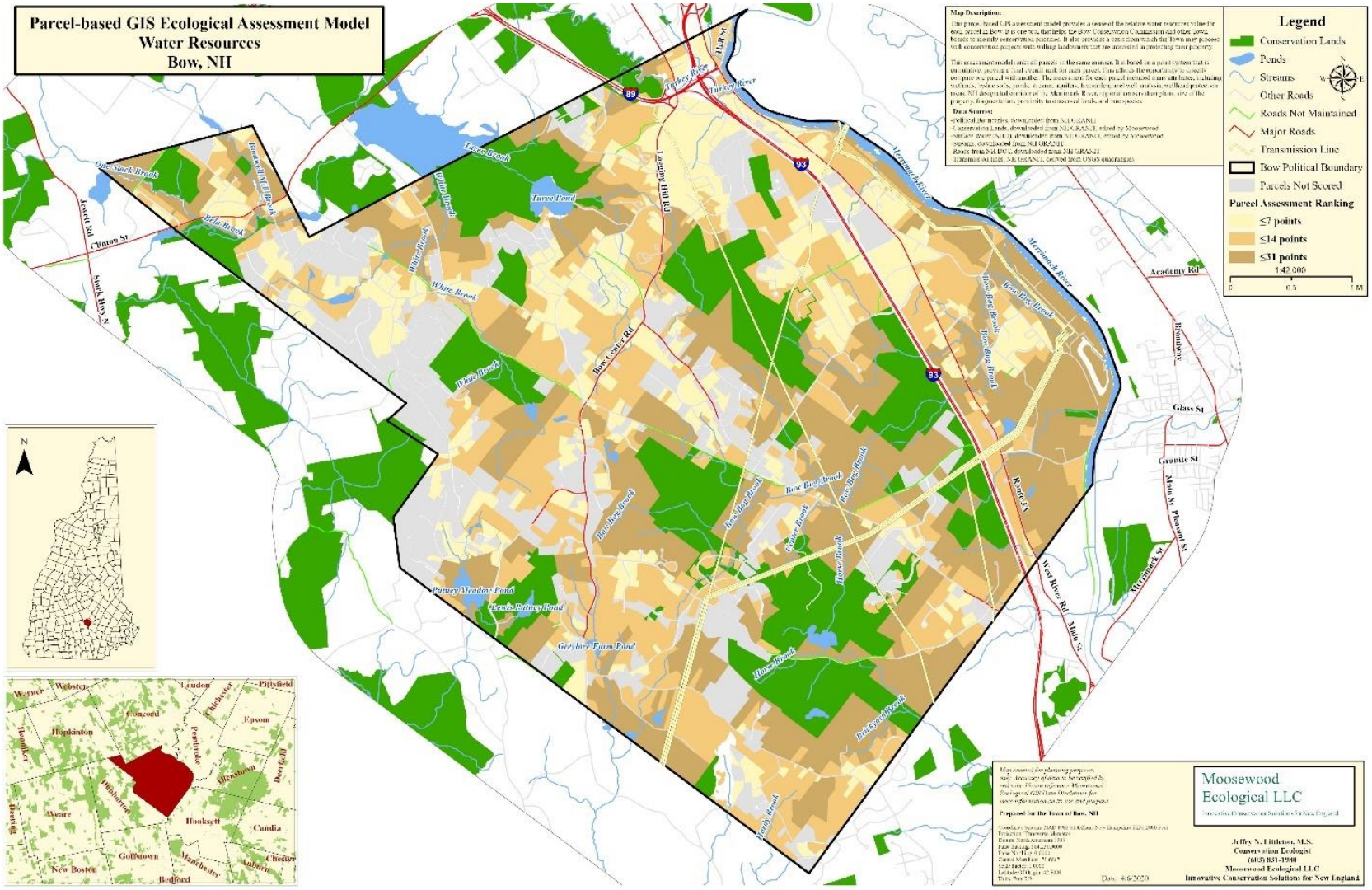
X. Rare species (data from Moosewood)

a. 1 species present	1 point
b. 2 species present	2 points
c. 3 or more species present	3 points

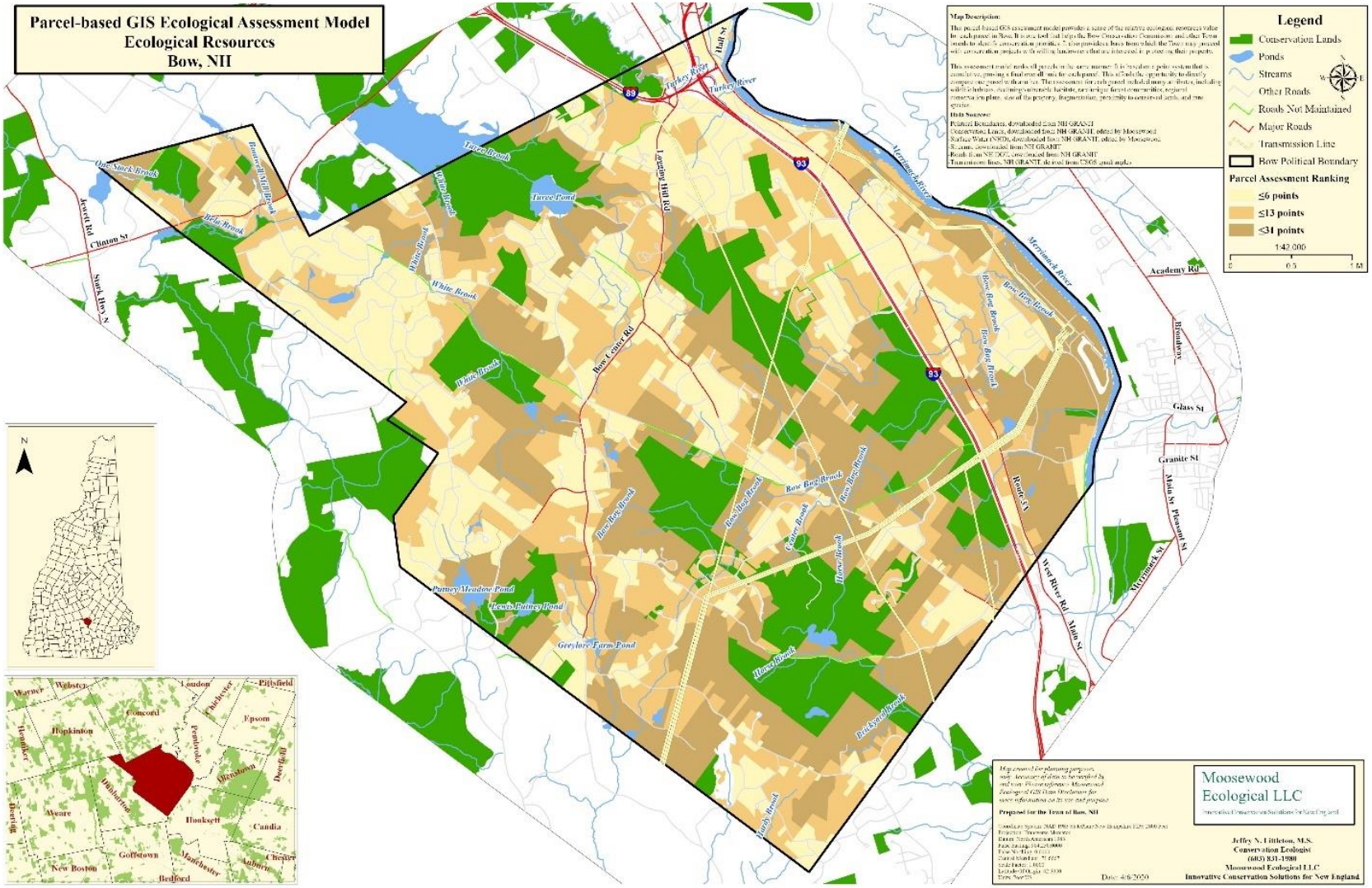
Appendix B: Agricultural and Forest Resources Co-occurrence Map

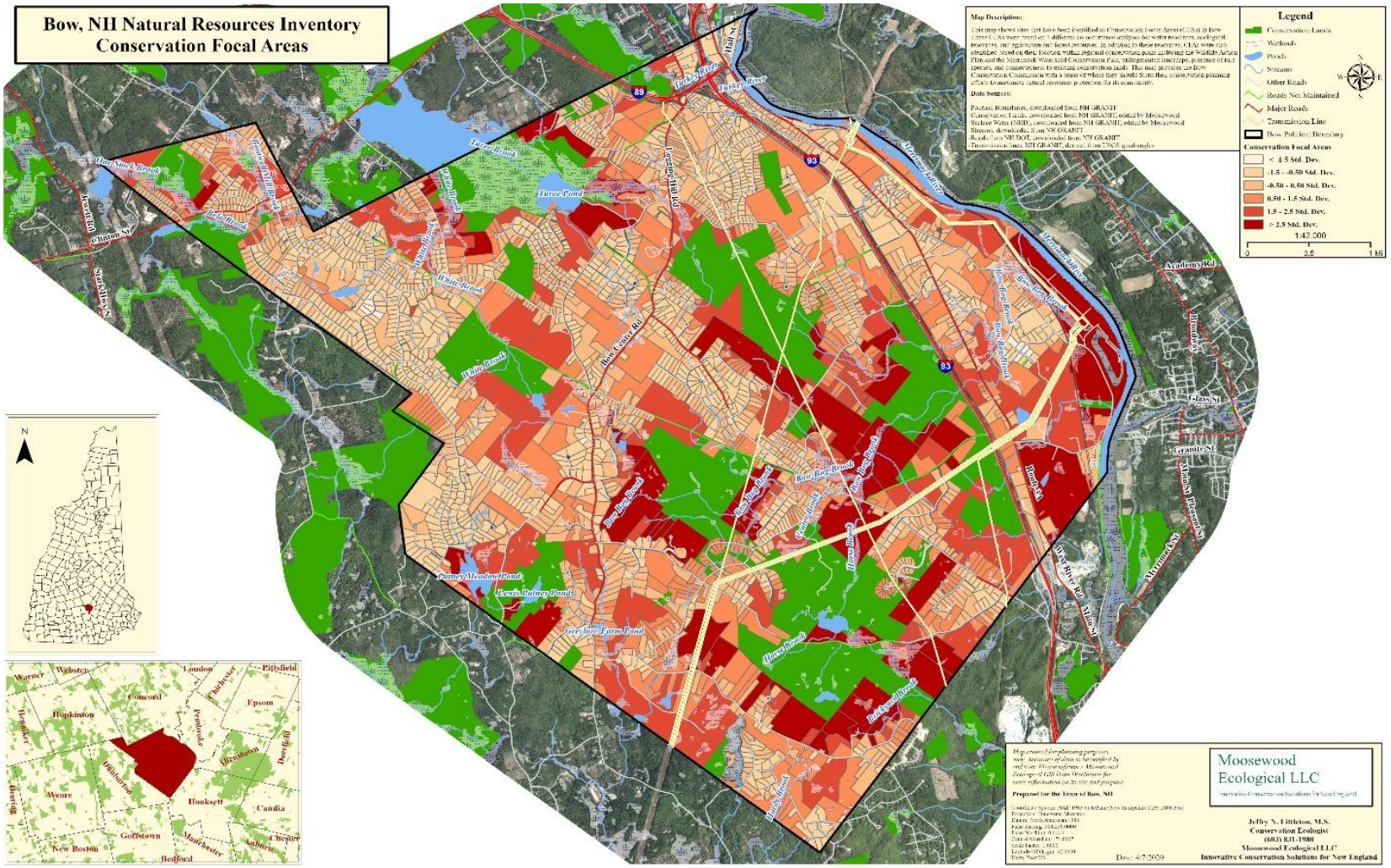


Appendix C: Water Resources Co-occurrence Map



Appendix D: Ecological Resources Co-occurrence Map



Appendix E: Conservation Focal Areas Map

CFA map for the town of Bow, NH. Values are reported as standard deviations from the mean in terms of conservation value. Red areas indicate areas of high conservation value according to the scoring criteria (std. dev. ≥ 2.5, mean=26.04).